Journal of Engineering Science and Technology
Special Issue on INCITEST2025 Vol. 21, No. 2 (2026) 97 - 104

IDENTITY-PRESERVING FACE RESTORATION FOR LOW-RESOLUTION CCTV IMAGES USING GAN AND STYLEGAN2

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Abstract

This research is a novel approach for face restoration that aims to preserve identity in low-resolution CCTV images. Our system uses a hybrid GAN-based architecture that integrates StyleGAN2 with a channel-split spatial feature transform (CS-SFT), a design that is crucial for balancing visual realism and identity fidelity. The research simulates synthetic images of real-world conditions by applying a multi-stage degradation removal process and extensive data augmentation, including motion blur, noise, and compression. Experiments were conducted using the FFHQ and CelebA datasets to evaluate performance using LPIPS, FID, NIQE, and PSNR metrics. The resulting model achieved competitive scores (LPIPS: 0.3320, FID: 59.75, NIQE: 4.491, PSNR: 24.40). We compared it with the PULSE and GPEN models to see the performance of the model that was built. In face recognition tests, this model achieved 93.55% accuracy on blurred images and 58.06% under full CCTV-effect degradations. The main contribution of this research is the integration of StyleGAN2 and CS-SFT into a dedicated pipeline for CCTV footage so that it can still produce perceptual image quality and identity preservation compared to existing methods.

Keywords: CCTV image enhancement, CS-SFT, Face restoration, GAN, StyleGAN2.

1. Introduction

Closed-circuit television (CCTV) has become a quite important technology in public, commercial, and residential security systems. However, cost limitations often result in the use of low-resolution cameras, which causes degraded image results. This can result in the hindered performance of more accurate face recognition [1, 2]. Facial recognition technology is widely implemented in automated surveillance systems to identify a person from stored databases, but its performance decreases significantly in poor-quality CCTV conditions [2-4].

Several cases of misidentification in the world often occur due to the existence of low-quality surveillance image data. One example that emerged is the Robert Williams case, where recognition errors resulted in mistaken arrest, where this was caused by ethical and technical problems surrounding the existing system [5]. These challenges indicate the need for image restoration techniques that can be improved, which are capable of improving facial clarity while preserving the identity characteristics of the facial image.

To overcome this problem, this research proposes an identity-preserving restoration model using a GAN-based framework to then be enhanced with the StyleGAN2 model and Channel-Split Spatial Feature Transform (CS-SFT). The goal is to improve perceptual image quality while maintaining identity integrity across various real-world CCTV degradations. Furthermore, the following discussion will discuss related previous research in face restoration and highlight existing research gaps.

2. Literature Review

Restoring facial images from degraded surveillance footage has long been a computer vision challenge. Early models like FaceNet perform well on high-quality data but degrade under severe CCTV-like conditions [6-8]. Deep learning-based recognition has also been developed for embedded object identification systems [9]. Generative adversarial networks (GANs) are widely used for image enhancement, with variants such as SRGAN, Real-ESRGAN, and GAN-based super-resolution models [10-15]. These models reconstruct realistic images from low-quality inputs using adversarial learning, including for surveillance imagery [16].

Blind face restoration has emerged as a specialized field with methods like GPEN and GFPGAN restoring high-quality faces but potentially causing identity shift. Advanced models such as VQFR, DR2, and CodeFormer improve stability through codebook or diffusion-based approaches, while BFRFormer employs multi-scale attention mechanisms. Recent work focuses on feature enrichment for image restoration [17-23].

Despite technological advances, many methods still rely on synthetic degradation assumptions and generalize poorly to real CCTV conditions like motion blur and compression artifacts [7, 8]. A model specifically designed to preserve identity under complex degradations is needed. This research addresses this gap by integrating StyleGAN2 with CS-SFT in a multi-stage pipeline, supported by CNN architectures like U-Net and efficient designs such as GhostNet [24-27].

3. Method

This research develops a multi-stage identity-preserving restoration framework optimized for real-world adapted CCTV conditions. The methodology conducted consists of four stages among them are: (i) data collection and augmentation, (ii) degradation removal, (iii) identity-preserving restoration, and (iv) performance evaluation. The entire process as seen in Fig 1.

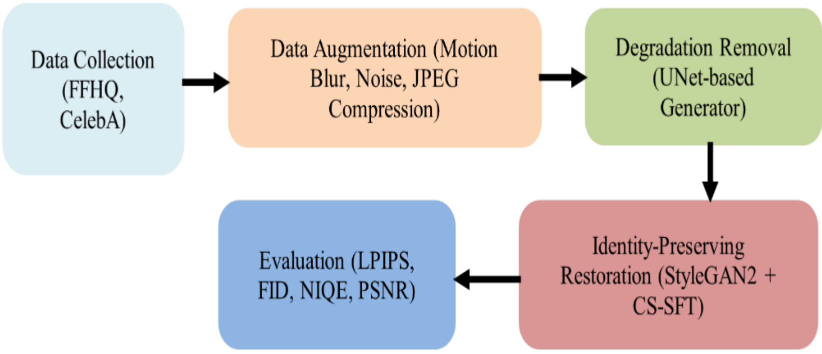

Fig. 1. Workflow of the developed system.

3.1. Data collection and augmentation

In this research, high-quality facial images were obtained from the FFHQ and CelebA datasets. To be able to simulate realistic CCTV conditions, various degradations were carried out including motion blur, Gaussian noise, JPEG compression, contrast reduction, and resolution downsampling. This augmentation process is very important to be able to reduce the gap between synthetic data and real-world CCTV surveillance data [7, 12].

3.2. Degradation removal (stage 1)

At this stage, the U-Net-based model generator performs initial restoration by separating latent identity representation from spatial structural details [25]. This step aims to remove major degradations such as blur and compression artifacts.

3.3. Identity-preserving restoration (stage 2)

The architectural efficiency is further supported by lightweight feature extraction modules inspired by GhostNet [27]. his multi-stage integration is aligned with other advanced reconstruction frameworks that combine domain knowledge with deep architectures, such as PID3Nt [28]. In this stage 2 part, the output from Stage 1 is fed into a StyleGAN2-based generator equipped with channel-split spatial feature transform (CS-SFT) [24]. Subsequently, the CS-SFT module modulates the internal features of StyleGAN2 using the generated spatial cues, enabling better retention of identity characteristics.

This architectural efficiency is further supported by lightweight feature extraction modules inspired by GhostNet [27]. This multi-stage integration aligns with other advanced reconstruction frameworks that combine domain knowledge with deep learning architectures, such as in the PID3Net model [28].

3.4. Loss functions

The model is optimized using a composite loss function combining reconstruction loss, perceptual loss, adversarial loss, and identity loss [18, 29], This approach maintains an optimal balance between visual realism and identity fidelity.

3.5. Evaluation metrics

Model performance is compared with state-of-the-art approaches - including PULSE, GPEN, VQFR, GFPGAN, CodeFormer, and DR2 - using LPIPS, FID, PSNR, NIQE, and identity recognition accuracy [17-19, 30].

4. Results and Discussion

Experiments were conducted using degraded FFHQ and CelebA images. Table 1 presents quantitative comparisons. The training process to produce the model was conducted in two phases as shown in Fig. 2. In Phase 1 (U-Net pre-training) showed a stable loss reduction. Meanwhile, Phase 2 (full GAN training) showed oscillations typical of adversarial learning but remained convergent, where the graph still indicated results moving toward better restoration quality.

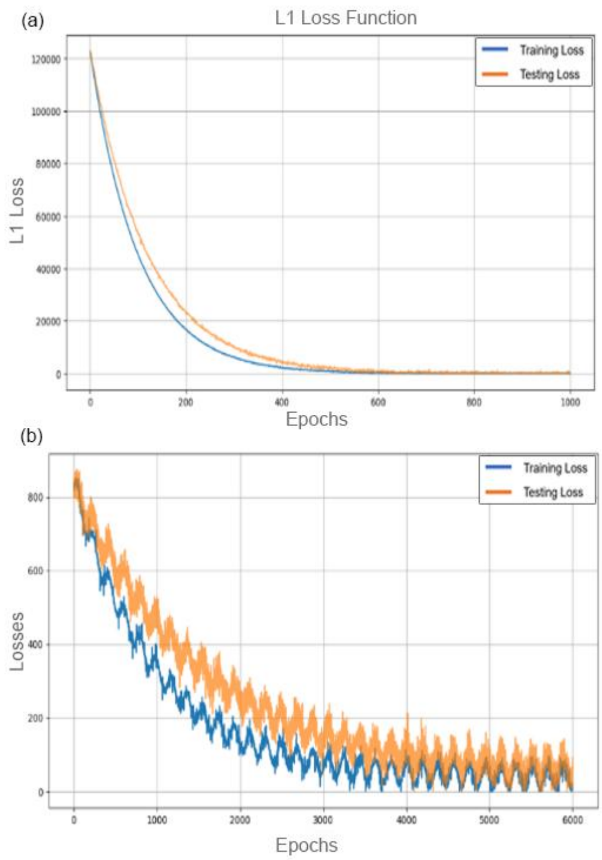

Fig. 2. (a) Train and test losses in the early phase (b) and train and test losses in the final phase.

Table 1. Results of evaluation.

Method	LPIPS↓	FID↓	NIQE↓	PSNR↑
Input	0.4830	145.22	12.771	25.35
PULSE	0.4851	67.56	5.305	21.61
GPEN	0.3746	63.52	4.577	24.28
VQFR	0.4311	66.34	3.688	21.93
GFPGAN	**0.3254**	**56.88**	4.021	23.54
CodeFormer	0.3451	71.21	4.432	23.38
DR2 + VQFR	0.3284	63.82	**3.481**	23.50
My-Model	0.3320	59.75	4.491	**24.40**

The developed model demonstrated competitive LPIPS and FID performance with an increased PSNR score, indicating improved pixel-level accuracy. Although GFPGAN achieved slightly higher perceptual scores, the proposed model offered superior identity retention. These findings confirm that the designed model is well-suited for both forensic applications and surveillance systems.

Figure 3 visually compares degradation removal results with restored images. These results demonstrate the model's capability to reconstruct important identity features while minimizing artifacts. Visual inspection shows that PULSE and GPEN tend to produce artificial textures, whereas our model maintains clearer facial details without altering key attributes such as eye shape and facial contours. This is crucial for mitigating misidentification risks.

Fig. 3. Comparison results: a) result of degradation removal, b) image restoration result.

Figure 4 provides visual examples demonstrating identity preservation, comparing the original input, degraded image, and restored result. Face recognition accuracy results are shown in Table 2. The results confirm that the proposed model preserves identity significantly better than prior methods.

Fig. 4. Example of identity preservation: (a) original image, (b) degraded image. (c) restored image by the proposed method.

Table 2. Evaluation result face recognition.

Input	Blurring	CCTV Effect	Method	Blurring Face Restoration	CCTV Effect Face Restoration
96.77%	83.87%	38.71%	DR2 + VQFR	0%	0%
			PULSE	0%	0%
			GPEN	90.32%	45.16%
			GFPGAN	93.55%	51.61%
			MyModel	**93.55%**	**58.06%**

5. Conclusion

Based on the testing and analysis results, the proposed identity-preserving face restoration model effectively enhances degraded CCTV image quality while maintaining the required identity fidelity. The two-stage GAN framework integrating U-Net, StyleGAN2, and CS-SFT demonstrates competitive performance across LPIPS, FID, NIQE, and PSNR metrics. Although perceptual quality remains marginally lower than top-performing models like GFPGAN and DR2, the proposed approach shows superior identity preservation. Future research directions include expanding dataset diversity, enhancing perceptual consistency under extreme degradation conditions, and developing lightweight versions suitable for real-time CCTV systems.

Acknowledgements

The authors would like to express their sincere gratitude to their research collaborators for their valuable contributions throughout this study. Special appreciation is extended to DP3M Universitas Komputer Indonesia (UNIKOM) for providing research support and guidance. The authors also acknowledge Universitas Komputer Indonesia (UNIKOM) for facilitating the research environment and resources that made this work possible.

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